

Swarup K Subudhi

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experience

Graduate Research Assistant, 2022-present

- **Development of Three-Dimensional High-Energy X-Ray Photoelectric Polarimeter Assembly Using Additive Manufacturing Technology**
Sponsor: NASA-Goddard Space Flight Center, Jan 2024 – present
- **Thermal Cycling Reliability of Printed Electronics**
Sponsor: Laboratory of Physical Sciences, June 2023 – Dec 2023
- **Reliability Validation and Sustainment of Direct-Write Printed RF Devices**
Sponsor: NextFlex, Jan 2022 – May 2023

Senior Manager, Manufacturing (Transmissions), 2019-20

Tata Motors Ltd. CVBU, Jamshedpur

- Improved Operational Line Efficiency of Gearbox Assembly Line by improving takt time & process quality parameters through process planning optimization
- Implemented low-cost ANDON analysis to capture assembly line losses through integration of IoT based tools with existing assembly line equipment & implemented loss-reduction plan

Assistant Manager, NPI (Transmissions), 2016-19

Tata Motors Ltd. CVBU, Jamshedpur

- Managed delivery of auxiliary gearboxes for 6X6 and 8X8 defense heavy trucks
- Launched ten new gearboxes (as per market requirements) and integrated into mass production through implementation of change-management processes

education

Ph.D., Mechanical Engineering (exp. May 2026)

University of Maryland, College Park, MD, USA

Dissertation: Magnetic Films & Drops: And their Interactions with Other Viscous Mediums

Courses: Mechanical Fundamentals of Electronic Components, Composite Materials, Interfacial Fluid Mechanics, Modelling Material Behavior, Probabilistic PoF & Accelerated Testing, etc.

Bachelors of Technology, Mechanical Engineering, 2016

National Institute of Technology, Rourkela, Odisha, India

skills

CAD: SolidWorks, Pro-E; **CAE/FEA:** ANSYS Mechanical, Workbench, Icepak, Abaqus CAE; **CFD:** Fluent, COMSOL; **CAM:** Autodesk 360, Solidworks CAM, nScryptPro; **Fabrication:** Laser Tooling, 3D Printing, Subtractive Machining, Mold Dev; **Metrology:** GD&T, SEM, Laser/ Optical Microscopy, Surface Profiling; **Material Characterization:** CTE, DMA (Stress Tests), DSC; **Rheology:** Density, Viscosity, Surface Tension, zeta-Potential, Size Distribution; **Control:** Arduino, PID, MATLAB GUI, Sensors; **Integration:** Prototyping, Circuit Design, Testing Matrix Development

Professional Tools: DFA/DFM, BOM, PLM, Six-Sigma, DoE, PSR, PFMEA, NPI Process

Computation & Analysis: Matlab & SimuLink (Optimization & Image Processing Toolbox), Mathematica, JupyterLab; **Reliability Testing:** Thermal Cycling, Temp-Humidity Bias, Vibration Loading, Deformational Loading; **Statistical Analysis:** Acceleration Factors, Failure Analysis, DoE, ANOVA, Minitab